CAPSTONE PROJECT

PROJECT TITLE

Presented By:

1. Student Name- Vineet Singh
2. College Name- KIET Group of Institutions
3. Department- CSE(AI ML)

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OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT

<u>Personalized Workout & Diet Planner with AI</u>

Problem

Most fitness applications provide generic workout routines and diet plans that fail to consider:

- Individual fitness goals
- Body measurements
- Dietary preferences
- Cultural food habits
- Available workout resources
- Budget constraints

As a result, students often receive recommendations that are impractical, difficult to follow, and less effective.

Need

There is a need for an AI-powered solution that generates personalized fitness and nutrition plans tailored to each user's unique requirements.

PROPOSED SOLUTION

AI Workout Planner

The proposed system is a web-based AI application that creates customized workout and diet plans based on user inputs.

User Inputs

- Age
- Gender
- Height
- Weight
- Fitness Goal
- Activity Level
- Diet Type
- Food Culture
- Budget
- Available Equipment
- Health Conditions

AI Outputs

- Personalized Workout Routine
- Customized Diet Plan
- Calorie Recommendations
- Health Tips
- Lifestyle Suggestions

SYSTEM APPROACH

Technologies Used

- **Frontend**
  - Streamlit
  - Interactive Dashboard
  - Responsive UI
- **Backend**
  - Python
  - Prompt Engineering
  - User Data Processing
- **AI Model**
  - Google Gemini API
  - Personalized Content Generation
- **Deployment**
  - GitHub Repository
  - Streamlit Community Cloud
- **Development Environment**
  - Visual Studio Code
  - Python Virtual Environment

ALGORITHM & DEPLOYMENT

- Working Algorithm

  Step 1 - Collect user fitness information.

  Step 2 - Validate input data.

  Step 3 - Generate structured AI prompt.

  Step 4 - Send request to Gemini AI.

  Step 5 - Generate workout recommendations

  Step 6 - Generate diet recommendations.

  Step 7 - Display personalized plan to user.

- Deployment Process

  VS Code → GitHub Repository → Streamlit Cloud → Live Web Application

RESULT

Project Result

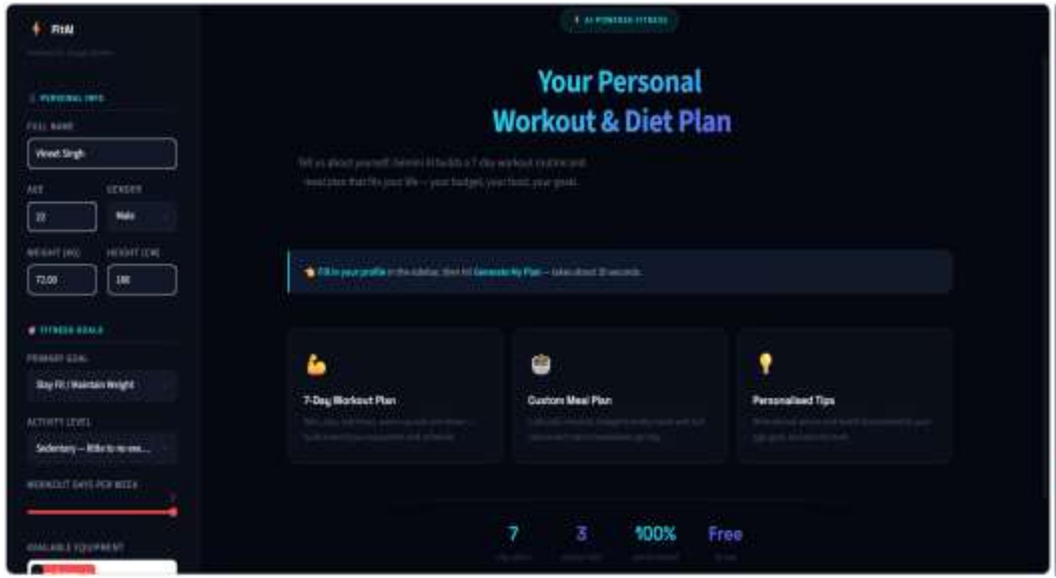
The developed application successfully generates personalized fitness and nutrition plans based on user requirements.

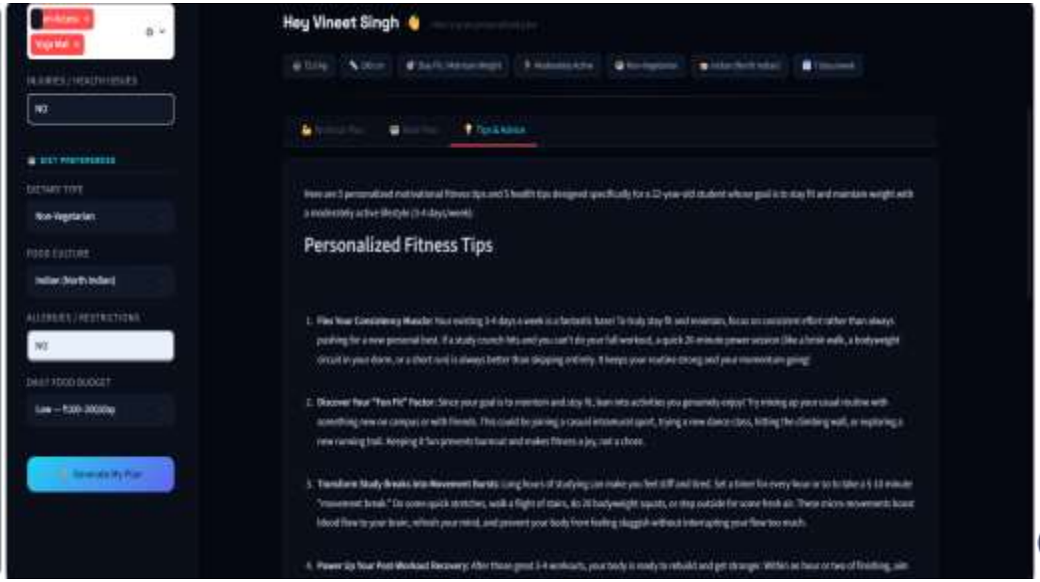
Achievements

- ✓ Personalized Workout Generation
- ✓ Customized Meal Planning
- ✓ AI-Based Recommendations
- ✓ User-Friendly Interface
- ✓ Cloud Deployment

Output

The application provides AI-generated workout and meal plans tailored to individual user requirements, ensuring personalized and effective fitness guidance.





CONCLUSION

Conclusion

- The AI-Powered Personalized Workout & Diet Planner successfully addresses the limitations of traditional fitness applications.
- By combining user-specific information with Generative AI, the system provides customized workout routines and meal plans that are practical, effective, and easy to follow.
- The project demonstrates how AI can improve health planning and create a better user experience through personalization.

FUTURE SCOPE

Future Scopes

- The system can be enhanced with:
- BMI & Body Fat Analysis
- Progress Tracking Dashboard
- Weekly Fitness Reports
- AI Fitness Chatbot
- Voice-Based Assistance
- Mobile Application
- Wearable Device Integration
- Real-Time Nutrition Tracking
- Multi-Language Support

These features can make the platform more intelligent and user-friendly.

REFERENCES

Hugging Face – The AI community building the future.

Streamlit • A faster way to build and share data apps

Welcome to Python.org

1. Supervised learning — scikit-learn 1.8.0 documentation

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GITHUB LINK

- Github Link : https://github.com/VineetSinghSF/AI_Workout_Planner
- Deployment Link: https://aiworkoutplanner-rrgpsncajy9agrvqhiewqd.streamlit.app/

THANK YOU